Reconstruction of Cloud-Covered Land Surface Temperature From Fengyun-4B TIR Remote Sensing Images Using an Improved DTC Model and Multisource Data Fusion Approach

Yan Li, Lin Zhu, Yi Lv, Yanhong Duan, and Xin Ye

***Abstract*—Land surface temperature (LST) is a critical parameter influencing energy and water exchanges between the land surface and atmosphere, and is used in diverse geoscience applications. However, thermal infrared remote sensing images are limited by cloud cover, hindering their all-weather applicability. This study proposes a novel algorithm for reconstructing cloud-covered LST by integrating FY-4B LST data with China Meteorological Administration Land Data Assimilation System (CLDAS) data. By constructing an improved diurnal temperature cycle (DTC) model named JNG06-STCI (JNG06 with spatiotemporally integrated interpolation based on critical LST initialization), this study improves the fitting performance of traditional JNG06 models in cloudy regions. On this basis, an XGBoost algorithm-based normalization model was developed to integrate DTC-fitted LST with reanalysis data, considering differences in energy balance under clear and cloud-covered conditions, thereby reconstructing cloud-covered LST. The results indicate that the JNG06-STCI model effectively mitigates the fitting bias of cloud-covered LST, with the improvement being most pronounced for areas subject to extensive and persistent cloud cover, thereby generating cloud-covered hypothetical clear-sky LST products with superior spatial continuity and a more physically plausible distribution. With ground validation from the Heihe Watershed Allied Telemetry Experimental Research (HiWATER) observation network in the Heihe River Basin, the integrated remote sensing-reanalysis product achieves a mean root-mean-square error (RMSE) of 3.52 K for cloud-covered LST reconstruction across six stations. This study develops an all-weather LST reconstruction framework using Chinese multisource spatiotemporal data, producing spatiotemporally continuous LST products under real-sky conditions. The resulting dataset supports fine-scale monitoring of land surface radiation and is applicable to climate change studies, drought monitoring, and urban thermal environment analysis.**

***Index Terms*—China Meteorological Administration Land Data Assimilation System (CLDAS), cloud reconstruction, diurnal temperature cycle (DTC), FY-4B, land surface temperature (LST).**

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I. Introduction

LAND surface temperature (LST) plays a pivotal role in land surface energy balance and serves as a key parameter for characterizing land–atmosphere interactions and energy exchange processes. It is widely applicable to diverse fields such as global climate change research, vegetation condition monitoring, urban heat island effect analysis, and disaster monitoring and early warning systems [1], [2], [3]. Thermal infrared remote sensing is a critical technique for efficiently acquiring LST data over large spatial extents, with advantages including continuous spatial coverage, broad observational scope, and high retrieval efficiency, and has thus been extensively adopted [4].

Over the years, significant progress has been made in the theory and methodology of LST retrieval from thermal infrared remote sensing, enabling the production of high-accuracy clear-sky LST products. However, cloud obstruction severely limits the ability of thermal infrared sensors to retrieve cloud-covered LST, resulting in spatially discontinuous LST datasets [5], [6], [7]. Consequently, effectively reconstructing missing cloud-covered LST values to generate spatially continuous, all-weather LST products is of significant importance, as it will enhance the spatiotemporal completeness and data availability of LST [8].

Methodologies for reconstructing all-weather LST can be categorized into two types according to the data employed: reconstruction based on valid observational data and multisource data integration.

Among the former, semi-empirical diurnal temperature cycle (DTC) models were among the earliest approaches and remain widely used. Based on the principle of energy exchange between the land surface and the atmosphere [9], these models employ variables implicitly related to LST—such as daytime LST amplitude, residual LST at sunrise, and atmospheric optical thickness [10]—to accurately capture the diurnal variation characteristics of LST under clear-sky conditions [11], [12]. The fitted diurnal curve estimates the hypothetical

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TABLE I
BAND PARAMETERS OF FY-4B AGRI

Band	Central Wavelength(μm)	Bandwidth(μm)	Spatial Resolution(km)	Band	Central Wavelength(μm)	Bandwidth(μm)	Spatial Resolution(km)
1	0.47	0.45~0.49	1	9	6.25	5.80~6.70	4
2	0.65	0.55~0.75	0.5	10	6.95	6.75~7.15	4
3	0.825	0.75~0.90	1	11	7.42	7.24~7.60	4
4	1.379	1.371~1.386	2	12	8.55	8.3~8.8	4
5	1.61	1.58~1.64	2	13	10.80	10.30~11.30	4
6	2.25	2.10~2.35	2	14	12.00	11.50~12.50	4
7	3.75	3.50~4.00(high)	2	15	13.3	13.00~13.60	4
8	3.75	3.50~4.00(low)	4				

clear-sky LST—the temperature that would occur without cloud cover—which differs from the actual cloud-affected LST. Classical representatives include the GOT01 [13], VAN06 [14], INA08 [15], JNG06 [16], and GOT09 [9]. These models have already been applied to fill gaps in satellite-derived LST products [17], [18]. However, when directly applied to cloudy conditions, DTC models yield hypothetical clear-sky LST estimates that deviate from actual cloud-covered LST because they do not account for changes in radiative forcing and surface energy balance caused by cloud cover [19]. The current study also reveals that due to the lack of effective observational data caused by cloud cover, existing DTC models exhibit limitations in fitting cloud-covered areas—they cannot accurately characterize the diurnal variation curve of LST at the pixel scale, and the resulting hypothetical clear-sky LST dataset shows pronounced spatial discontinuity.

Reanalysis data constitute a vital source for reconstructing cloud-covered LST. They are generated through data assimilation technology, which integrates satellite and ground-based observations into a unified land surface process model [20], [21], [22]. This data assimilation approach simulates the surface energy balance processes in cloud-covered areas and captures the variation trends of LST in predominantly cloud-covered regions. Therefore, integrating thermal infrared remote sensing data with reanalysis data—a multisource data integration algorithm—can compensate for the DTC model's neglect of cloud-induced energy balance alterations [23], [24]. However, challenges such as the coarse spatial resolution and inherent uncertainties of reanalysis data must be addressed during integration [7], [25], [26].

FY-4B is the first operational satellite of China's new-generation geostationary meteorological satellite series. Its deployment marks the operational transition of China's next-generation geostationary observation system [27]. China Meteorological Administration Land Data Assimilation System (CLDAS), independently developed by the China Meteorological Administration (CMA), entered operational use in early 2022. Evaluations by the National Climate Center indicate that CLDAS generally outperforms commonly used NCEP reanalysis products in terms of data quality, timeliness, and stability [28]. Further validation studies have demonstrated that, compared to similar international products, CLDAS achieves higher accuracy and a more physically reasonable spatiotemporal distribution over China [29], [30], [31]. Despite this, the majority of existing reanalysis studies are still based on international datasets like ERA5 and NCEP [32], [33], [34], and the potential of CLDAS data has not yet been fully exploited.

Several two-step frameworks have been proposed for all-weather LST reconstruction, typically combining an initial temporal gap-filling step with a subsequent reanalysis-based correction. Zeng et al. [35] applied a harmonic analysis of time series (HANTSs) to reconstruct cloud-contaminated MODIS LST and then corrected for cloud radiative effects using CFSv2 reanalysis data. Jia et al. [36] fit a DTC model to GOES-R ABI clear-sky observations and merged the results with ERA5 to produce a 2-km hourly all-sky product. Zhang et al. [37] combined DTC-fitted clear-sky LST with reanalysis-based correction over MODIS data across multiple surface types. A shared limitation of these approaches is that the temporal fitting step is assumed to remain reliable even when extensive cloud cover removes observations at critical diurnal phases (e.g., sunrise and daily peak), a condition that may compromise fitting accuracy, as we demonstrate in Section III-C. Liu et al. [38] bypassed the DTC step by directly coupling FY-4A/Advanced Geostationary Radiation Imager (AGRI) radiance with reanalysis-derived energy fluxes through surface energy balance theory; however, because the satellite–reanalysis bias was handled as a single correction term, the approach could not isolate the systematic interdataset offset from the cloud-driven energy-balance perturbation. Across these studies, a second common gap persists: the bias between products is corrected as one lumped quantity, without decomposing it into a clear-sky systematic component and a cloud-induced component that arises from distinct physical mechanisms.

In light of these considerations, this study proposes a novel algorithm for reconstructing cloud-covered LST by developing

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a normalized satellite-reanalysis data fusion model. The key contributions of this work are threefold as follows.

1) The DTC fitting degradation problem is addressed. Existing two-step frameworks implicitly assume that DTC models can reliably estimate hypothetical clear-sky LST given sufficient surrounding observations. This study shows that missing observations at critical time points (sunrise and daily peak) may cause fitting distortions and pronounced spatial discontinuity in cloud-covered pixels. The JNG06-STCI model (JNG06 with spatiotemporally integrated inter polation based on critical LST initialization), which incorporates hierarchical spatiotemporal collaborative interpolation, is proposed to constrain the diurnal and spatial trends of cloud-covered LST and ensure spatial coherence of the hypothetical clear-sky baseline.
2) A new normalization framework integrating satellite and reanalysis data is developed. The satellite-reanalysis bias is explicitly decomposed into two physically interpretable terms: a systematic interdataset discrepancy calibrated under clear-sky conditions using FY-4B as the reference, and a cloud-induced energy balance deviation derived from CLDAS cloudy-versus-clear differences. Each term is modeled with independent training data and distinct physical drivers, rather than being lumped into a single undifferentiated correction.
3) An hourly all-weather LST product at 4-km spatial resolution is generated by leveraging the high temporal density of FY-4B observations (15-min intervals) together with the demonstrated accuracy of CLDAS reanalysis over the study region, thereby filling the cloud-induced gaps that conventional thermal infrared products cannot resolve.

The main findings are summarized as follows: the JNG06-STCI model reduces the maximum cloud-covered fitting deviation from over 14 to within ~1 K and lowers all scene-level spatial-smoothness metrics by 32.8%–60.1% relative to JNG06, restoring spatial continuity consistent with surrounding clear-sky observations; the CLDAS-based energy-balance correction compensates for the cloud-induced bias that the DTC baseline cannot capture, reducing the station-level bias from up to ~4.5 to within ~1.5 K; and the fused product, validated against six Heihe Watershed Allied Telemetry Experimental Research (HiWATER) stations, achieves a mean root-mean-square error (RMSE) of 3.52 K, approaching the clear-sky accuracy of 2.99 K and outperforming the standalone JNG06-STCI (6.68 K) and raw CLDAS (4.53 K), with stable performance across stations (3.24–3.79 K).

II. Study Area and Data Sources

A. Study Area and Study Period

This study is conducted in the Heihe River Basin, which is located in the arid and semi-arid regions of western China, spanning the geographical range of 97.1°E–102.0°E and 37.7°N–42.7°N with a drainage area of 143 000 km^2 [39], as shown in Fig. 1. The land cover within the basin is classified based on the International Geosphere-Biosphere Program (IGBP) system, derived from the MODIS Band 1 product

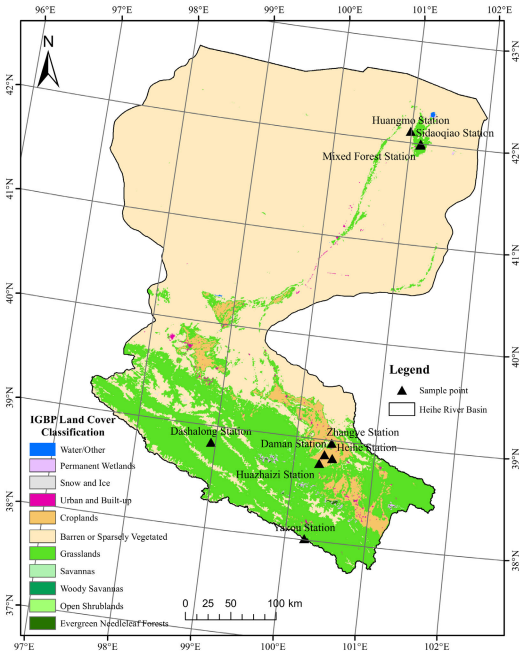


Fig. 1. Study area: IGBP land cover distribution of the Heihe River Basin (97.1°E–102.0°E, 37.7°N–42.7°N). Inset: location within China. Triangles denote the six HiWATER validation stations.

[40]. As the second-largest inland river basin in China, the Heihe River Basin features distinct landscape zonality. Linked by the river system, the basin exhibits a complete landscape gradient from upstream to downstream: ice/snow and frozen ground, forest, grassland, oasis (farmland and riparian forest), desert, and terminal lake. The basin encompasses both cold regions and arid zones, where the mountain cryosphere and the extremely arid river terminal areas present a striking contrast, making it an ideal site for carrying out land surface process observation and scientific research [41]. The data used in this study, all from the year 2023, were acquired over the following periods. The FY-4B AGRI satellite imagery and the CLDAS reanalysis fields span two full calendar months, October and December: the entire October record is used for model development and parameter tuning, while the entire December record is reserved for independent validation. The IGBP land cover classification is the MODIS-derived product for the same year, and the HiWATER in situ measurements used for validation cover the full month of December.

B. Remote Sensing Data

The satellite data used in this study are derived from the AGRI onboard the FY-4B satellite. Equipped with 15 spectral bands (detailed parameters shown in Table I), AGRI provides full-disk images with a spatial resolution of 4 km and a temporal resolution of 15 min. These images, along with the official longitude and latitude lookup tables and data row-column index files for geometric correction, were obtained from the Fengyun Satellite Remote Sensing Data Service Network [42].

Among the satellite data, the LST product serves as the primary data source. In addition, near-infrared band reflectance

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TABLE II
INFORMATION OF VALIDATION STATION

Station	Longitude (E)	Latitude (N)	Landscape	IGBP Land Cover Classification
Desert Station	100.99	42.11	Reaumuria desert	Barren
Sidaoqiao Station	101.14	42.00	Tamarix	Open Shrublands
Daman Station	100.37	38.86	Maize	Croplands
Heihe Remote Sensing Station	100.48	38.83	Grassland	Grasslands
Huazhaizi Desert Steppe Station	100.32	38.77	Kalidium foliatum desert	Barren
Zhangye Wetland Station	100.45	38.98	Reed	Permanent Wetlands

(NIR), red band reflectance (Red), and net shortwave radiation (NSR) are used as auxiliary data for multisource data fusion. The Red and NIR are employed to calculate the normalized difference vegetation index (NDVI), which can reflect surface vegetation coverage and land cover types. Given its correlation with LST [7], NDVI is applicable for selecting similar pixels in spatiotemporal collaborative interpolation and feature modeling in multisource fusion processes. As a core parameter influencing surface energy balance, NSR can characterize the solar radiation input received by the land surface, providing a basis for correcting energy balance biases in cloud-covered areas.

C. Reanalysis Data

This study employed the LST assimilation dataset of the CLDAS Version 2.0 (CLDAS-V2.0). Developed by the National Meteorological Information Center of CMA, CLDAS-V2.0 covers the Asian region (0 ~ 65°N, 60 ~ 160°E). It provides a long-term series of equal latitude-longitude grid fusion analysis products with a spatial resolution of 0.0625° and a temporal resolution of 1 h, including five categories: atmospheric forcing field products, LST analysis products, soil moisture products, soil temperature analysis products, and soil relative humidity analysis products. The CLDAS-V2.0 dataset is developed by integrating multisource observational data (including ground observations from over 2400 national automatic weather stations and nearly 40 000 regional automatic weather stations, ECMWF and GFS numerical forecast products, satellite precipitation products, etc.), and employs technologies such as multiscale 3-D variational analysis system (STMAS), optimal interpolation (OI), cumulative distribution function (cdf) matching, physical inversion and topographic correction [28], [43], [44]. In this study, the CLDAS LST data were used to correct the cloud-induced energy balance bias of the DTC model caused by neglecting the energy balance under cloud cover. Since the native spatial resolution of CLDAS (0.0625°) is coarser than that of FY-4B AGRI (4 km), the CLDAS LST grids were resampled to the FY-4B 4-km grid using bilinear interpolation to enable pixel-to-pixel fusion.

D. Validation Data

The ground-based validation data were obtained from the HiWATER observation network [45]. This observation network consists of three super stations and 18 ordinary stations, covering the upstream, midstream, and downstream areas of the Heihe River Basin. The station types cover the main land surface types within the basin, including grassland, farmland, wetland, and desert. In this study, the optimized observation network data of 2023 were selected. At each site, the corrected upwelling and downwelling longwave radiation measured by four-component net radiometers, together with the surface broadband emissivity derived from the ASTER Global Emissivity Dataset (ASTER GED) for removing the reflected atmospheric downwelling longwave component, were used to retrieve the in situ LST through the Stefan–Boltzmann law [46], [47]. The in situ measurement data from the HiWATER observation network have been widely used in data validation efforts [48], [49], [50], and can provide a reliable ground truth reference for validating the accuracy of the FY-4B LST product and the cloud-covered LST reconstruction results in this study. The information of the sites used for validation is presented in Table II.

III. METHODOLOGY

A. Workflow

The workflow of this study is illustrated in Fig. 2. The algorithm is organized into two complementary steps. In Step 1, the JNG06-STCI model fits a spatiotemporally continuous clear-sky LST sequence from FY-4B observations; for cloud-covered pixels, this sequence represents the hypothetical clear-sky baseline—the LST that would have occurred without clouds—rather than the actual cloud-covered LST. In Step 2, the CLDAS reanalysis data are fused with this baseline through a normalization procedure that first removes the systematic discrepancy between the two datasets under clear-sky conditions and then corrects the cloud-induced energy balance deviation, converting the hypothetical clear-sky baseline into a realistic cloud-covered LST estimate. The final all-weather product combines the observed clear-sky LST with the reconstructed cloud-covered LST, yielding an hourly, 4-km resolution dataset.

B. Construction of LST Diurnal Cycle Model

Under clear-sky conditions, comparative performance evaluations of various DTC models have shown that the JNG06 and GOT09 models exhibit the highest fitting accuracy, with an

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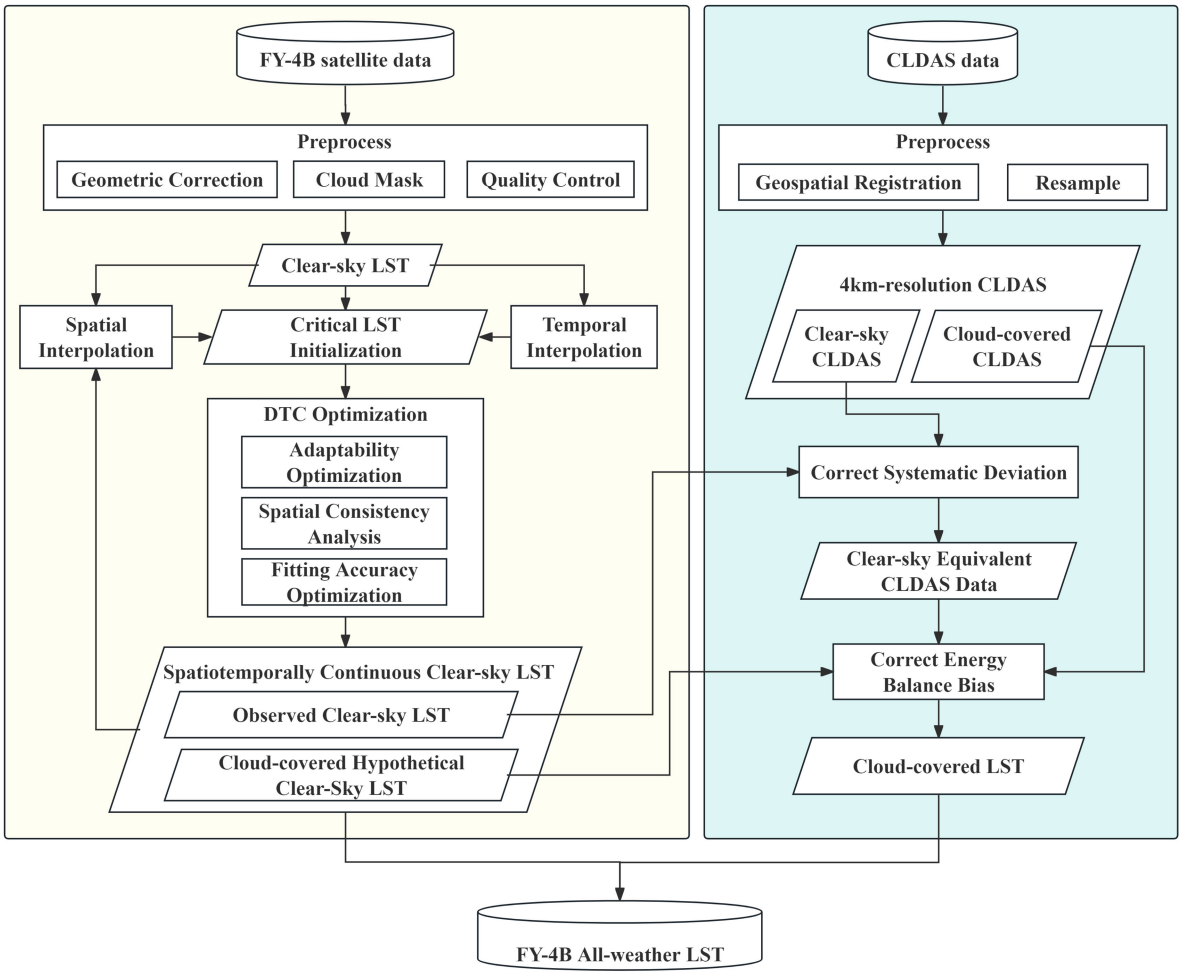


Fig. 2. Workflow for reconstructing all-weather LST dataset based on the optimized DTC model and multisource data fusion algorithm.

overall RMSE of approximately 0.5 K, outperforming models such as GOT01, VAN06, and INA08, which have an RMSE of around 0.8 K [51]. This conclusion has been further validated in studies conducted in China, where the RMSE values for the JNG06 and GOT09 models were 0.89 and 0.92 K [10]. However, the GOT09 model relies on parameters such as atmospheric optical thickness, resulting in a relatively complex modeling process, while its fitting accuracy does not show a clear advantage. Considering these two factors, this study selects the JNG06 model as the foundational model.

The JNG06 model, proposed by Jiang et al. [16], employs a cosine function combined with an exponential decay function to simulate diurnal and nocturnal LST variations, respectively, while ensuring the continuity of LST and its first derivative at the transition moment.

The expression of the JNG06 model is as follows:

$$\begin{cases} T_1(t) = T_0 + T_a \cos\left(\beta\left(t - t_m\right)\right), & t < t_s \\ T_2(t) = b_1 + b_2 e^{\alpha(t-t_s)}, & t \geq t_s. \end{cases} \tag{1}$$

The auxiliary parameter satisfies

$$\begin{cases} b_1 = T_0 + T_a \cos(\beta(t_s - t_m)) - b_2 \\ b_2 = \dfrac{[-\beta T_a \sin(\beta(t_s - t_m))]}{\alpha}. \end{cases} \tag{2}$$

The six free parameters included in the model's parameter system $(T_0, T_a, t_m, t_s, \beta, \alpha)$, along with their physical meanings and methods for determining initial values, are presented in Table III.

C. Spatiotemporal Integrated Optimization of LST Interpolation

The JNG06 model incorporates six free parameters. Theoretically, when a target pixel has more than six valid observations within a single diurnal variation cycle, the clear-sky LST at any given moment can be estimated. However, data gaps caused by cloud occlusion degrade the fitting accuracy of the JNG06 model. In practical applications, this study found

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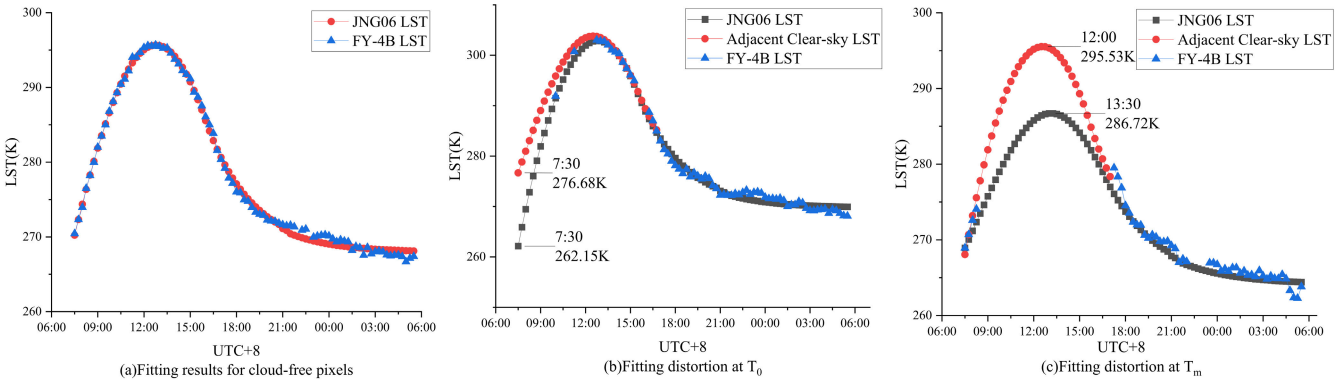


Fig. 3. JNG06 fitting results for three representative pixels. (a) All-day clear-sky. (b) Sunrise observations missing. (c) Peak-period observations missing. Blue: FY-4B clear-sky observations; red: JNG06 fit of the target pixel; and gray: fits of neighboring clear-sky pixels.

TABLE III
KEY PARAMETERS OF THE JNG06 MODEL

Parameter	Physical Meaning	Determination Method
T_0	Sunrise LST	First valid observation after sunrise.
T_a	LST Amplitude	Maximum valid LST - Minimum LST
t_m	Time of LST peak occurrence	$\frac{(T_{sr}+T_{ss})}{2}+1$
t_s	Start time of free LST decay	$T_{ss}-1$
β	Frequency parameters of daytime warming curve	$\frac{\omega}{\pi}$
α	Nighttime cooling rate parameters	Empirical Value: -0.3

In the table, T_{sr} and T_{ss} represent sunrise and sunset times, respectively, determined by the sunrise equation based on local latitude and longitude.

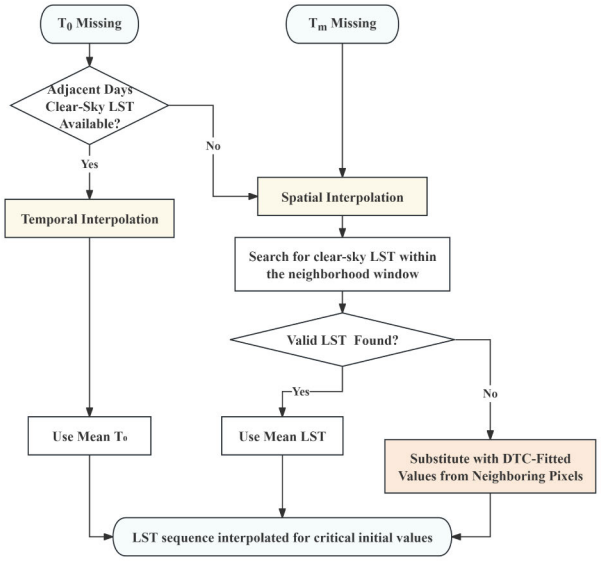


Fig. 4. Flowchart of the spatiotemporally collaborative interpolation optimization method.

that the model is highly sensitive to the integrity of two critical datasets: the sunrise-time LST (T_0) and the LST data corresponding to the peak period (T_m). The absence of data during these two periods will induce systematic deviations in the model inversion results.

Given the spatial continuity of LST under clear-sky conditions, the hypothetical clear-sky LST fitted by the DTC model for cloud-covered areas is expected to be spatially consistent with the actual clear-sky LST of neighboring pixels. Therefore, the difference between them can be used to evaluate the accuracy of the DTC model in fitting hypothetical clear-sky LST under clouds. Fig. 3 presents the fitting results of the JNG06 model. Under clear-sky conditions throughout the day, the LST dataset remains complete, as shown in Fig. 3(a), with the average RMSE across the day ranging only from 0.15 to 0.68 K. In Fig. 3(b), valid observation data for the target pixel during the sunrise period are missing, leading to the maximum discrepancy in the fitted sunrise LST (T_0), which differs from that of adjacent clear-sky pixels by up to 14.53 K, with a mean RMSE across all sampled pixels of 7.01 K. In Fig. 3(c), valid observation data for the target pixel during the LST peak period are missing. Comparative analysis with the fitting results of neighboring clear-sky pixels reveals discrepancies between the DTC fitting results of the target pixel and those of the all-day clear-sky pixels: the occurrence time of the LST peak (t_m) is shifted; the maximum difference in the LST peak value (T_m) reaches 8.81 K, with a mean RMSE across all sampled pixels of 4.53 K.

The results indicate that the traditional JNG06 model has inherent limitations in fitting LST for cloud-covered pixels. In cases where data during the T_0 or T_m periods are missing, the fitting results may be distorted, leading to inconsistencies with the spatial distribution characteristics of LST in surrounding pixels and undermining the spatial continuity of LST.

In response to the above issues, this study proposes a spatiotemporally collaborative interpolation optimization method. Building on the JNG06 model, we develop the JNG06-STCI model. By filling in the missing T_0 and T_m data and constrain-

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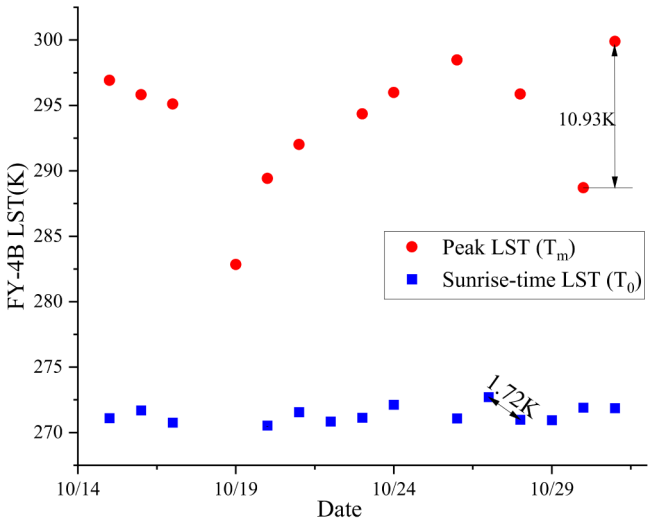


Fig. 5. Temporal evolution of the sunrise LST and the daily peak LST over 15 consecutive clear-sky days in late October 2023.

ing the spatiotemporal distribution characteristics of the DTC curve, the model enhances fitting accuracy for cloud-covered areas. The flowchart of the proposed method is shown in Fig. 4, and a summary of the specific methodology is provided as follows.

1) Temporal Interpolation (Gap-Filling for T_0): As shown in Fig. 5, analysis of 15-day observation data from late October in the study area reveals that under clear-sky conditions, the sunrise LST exhibits strong consistency between consecutive days. The interdaily deviation of LST at a given time can be quantified as

$$\delta T = \max_{i\in\{-1,+1\}} |T(t) - T(t+i)| \tag{3}$$

where $T(t)$ represents the LST at a specific phase (e.g., sunrise or daily peak) on a given day.

For sunrise LST (T_0), the observed δT is small, with an average of 0.92 K and a maximum of 1.72 K. Therefore, the sunrise LST for the target date can be reliably estimated as the mean of clear-sky sunrise LST from adjacent days

$$T_0' = \frac{1}{N}\sum T_{0,i}^{NearDate}, \quad N \geq 1 \tag{4}$$

where $T_{0,i}$ denotes the observed sunrise LST on a nearby clear-sky day, and N is the number of valid observation days.

In contrast, for the daily peak LST (T_m), the maximum δT exceeds 10 K, reflecting its high sensitivity to synoptic conditions and the poor representativeness of adjacent-day data. Hence, temporal interpolation is not appropriate for T_m, and spatial interpolation is employed to fill its missing values.

2) Spatial Interpolation (Gap-Filling for T_m): The spatial interpolation method fills missing LST values in a target pixel with valid observations from nearby similar clear-sky pixels at the same time. The NDVI exhibits a statistical relationship with LST and has been widely used as an environmental factor in LST downscaling studies due to its influence on surface thermal conditions [7], [52], [53]. Given the spatial continuity of LST, this study assumes that within a certain spatial range, differences in LST are primarily driven by differences in NDVI—i.e., pixels with similar NDVI values tend to have similar LST values. The spatial interpolation procedure is implemented as follows.

1) Check whether LST data at the peak LST period are missing for the target pixel.
2) Centered on the target pixel, search within a neighboring window for clear-sky pixels with similar NDVI values that have valid LST observations.
3) Assign the mean of these valid LST values to the missing pixel. The calculation is expressed as

$$T_{fill} = \frac{1}{M}\sum T_j^{clear}, \quad M \geq 1. \tag{5}$$

where T_j^{clear} denotes the observed LST of a similar clear-sky pixel at the target time, and M is the number of valid pixels within the search window.

3) Spatiotemporal Joint Interpolation: To address special cases in which either temporal or spatial interpolation alone fails—such as consecutive cloudy days or extensive thick cloud cover—a hierarchical joint gap-filling strategy was designed.

To fill in T_0, temporal interpolation is applied first. If sunrise periods on adjacent days are also cloud-obscured and no valid reference data are available, the method switches to spatial interpolation to estimate the initial sunrise LST.

When cloud cover is spatially extensive, pixels near the center of the cloud-covered area may have no valid LST observations within the search window, leading to the failure of spatial interpolation. Exploiting the spatial continuity of LST, the method then uses DTC-fitted LST from neighboring pixels as substitutes.

D. Normalization of Reanalysis and Remote Sensing Data Based on the XGBoost Algorithm

XGBoost is an efficient, flexible, and scalable machine learning algorithm based on the gradient boosting framework [54]. It improves prediction performance by integrating multiple decision trees and is widely applied in classification, regression, and ranking tasks. In contrast to traditional linear regression models, XGBoost can model complex, nonlinear dependencies [55], thereby enabling it to capture the intricate relationships and interactions between feature factors and LST.

In this study, an XGBoost-based normalization model was constructed to harmonize FY-4B satellite DTC-fitted LST data and CLDAS LST. Two physically distinct sources of bias exist between these heterogeneous datasets and need to be corrected separately, rather than being lumped into a single correction term as follows.

1) *Systematic Interdataset Discrepancy:* FY-4B LST is retrieved instantaneously from thermal infrared radiance via a split-window algorithm [56], whereas CLDAS LST is generated through multisource data assimilation driven by a land surface model [44]. Even under clear-sky conditions, where both datasets describe the same physical state, systematic differences persist due to sensor calibration, emissivity parameterization, atmospheric correction assumptions, and land surface model physics

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[57]. This bias is independent of cloud presence and must be calibrated using clear-sky pixel pairs as reference data.

2) *Cloud-Induced Energy Balance Deviation:* The DTC-fitted LST represents a hypothetical clear-sky baseline that ignores cloud-induced alterations in the surface energy balance, while CLDAS LST inherently captures such effects through its assimilation scheme. The difference between cloudy and clear-sky CLDAS LST therefore provides a physically grounded estimate of the energy balance correction required for the DTC baseline.

Accordingly, the fusion workflow is organized into three steps, with the first two targeting the two bias sources identified above and the third combining their results as follows.

1) *Systematic Bias Correction:* Under clear-sky conditions, the LST difference function between FY-4B and CLDAS data is established as

$$\Delta T_{clear}(x,t) = T_{clear}^{CLDAS}(x,t) - T_{clear}^{FY4B}(x,t) \quad (6)$$

$$\Delta T_{clear}(x,t) = f_{clear}(S(x,t)) + \varepsilon(x,t). \quad (7)$$

where $S = [NDVI, NSR]^T$ is the vector of difference-driving factors, $f_{\text{clear}}(\cdot)$ is the systematic difference term modeled by XGBoost, $\varepsilon \sim \text{N}(0, \sigma^2)$ spatially autocorrelated noise, whose covariance matrix is estimated via the variogram

$$\Gamma(h) = \frac{1}{2N(h)} \times \sum\nolimits_{|x_i - x_j| = h} (\varepsilon(x_i) - \varepsilon(x_j))^2. \quad (8)$$

2) *Energy Balance Bias Correction:* For cloud-covered pixels, the hypothetical clear-sky LST in the CLDAS reference frame, $T_{cloud_clear}^{CLDAS}$, is first obtained by transforming the JNG06-STCI hypothetical clear-sky LST $T_{cloud_clear}^{FY4B}$ with the systematic bias term f_{clear}. The energy-balance deviation ΔT_{SEB}, i.e., the LST difference induced by cloud-related energy balance variations, is then defined as the difference between the actual cloudy CLDAS LST T_{cloud}^{CLDAS} and this baseline, and is modeled as a function f_{SEB} of the driving factors plus a residual term ε

$$T_{cloud_clear}^{CLDAS}(x,t) = T_{cloud_clear}^{FY4B} + f_{clear}[S(x,t)] \quad (9)$$

$$\Delta T_{SEB}(x,t) = T_{cloud}^{CLDAS}(x,t) - T_{cloud_clear}^{CLDAS}(x,t)] \quad (10)$$

$$\Delta T_{SEB}(x,t) = f_{SEB}[S(x,t)] + \varepsilon(x,t) \quad (11)$$

where $S = [NDVI, NSR, t_{cloud}]^T$, and t_{cloud} is the duration of cloud coverage for the pixel.

3) *Actual Cloud-Covered LST Reconstruction:* Based on hypothetical clear-sky LST $T_{cloud_clear}^{FY4B}(x,t)$, and the energy balance correction term $f_{SEB}[S(x,t)]$, actual cloud-covered LST $T_{cloud}^{FY4B}(x,t)$ is reconstructed as

$$T_{cloud}^{FY4B}(x,t) = T_{cloud_clear}^{FY4B}(x,t) + f_{SEB}[S(x,t)]. \quad (12)$$

E. Model Accuracy Validation

This study validates the proposed algorithm from three aspects: the improvement effect of the JNG06-STCI model, the overall fitting accuracy of the DTC model under clear-sky conditions, and the accuracy of the reconstructed all-weather LST product.

The spatially seamless LST datasets fitted by the DTC model before and after improvement are analyzed. In the temporal dimension, the LST differences at sunrise and noon before and after the model improvement are compared to clarify the effect of the spatiotemporal interpolation method on constraining the shape of the LST curve and correcting fitting distortion. In the spatial dimension, based on the assumption that the hypothetical clear-sky LST of cloud-covered pixels has consistent spatial continuity with the LST of adjacent clear-sky pixels, we compare and analyze the spatial continuity of LST images between the JNG06 model and the improved JNG06-STCI model to validate the accuracy of the hypothetical clear-sky LST of cloud-covered pixels and verify the effectiveness of the model improvement.

Under clear-sky conditions, grouped validation is performed according to major land surface types (e.g., wetlands, grasslands, and deserts) and four different time periods (sunrise warming period, LST peak period, sunset cooling period, and free attenuation period). This is to validate the fitting accuracy of the model, as well as its fitting accuracy and stability under different underlying surfaces and time periods.

Under cloud-covered conditions, the in situ observation data from HiWATER ground stations in December 2023 are used as the reference true LST values. These observed data are compared with: 1) the hypothetical clear-sky LST fitted by the JNG06-STCI model; 2) the original CLDAS LST data; and 3) the reconstructed cloud-covered LST generated by the proposed multisource data fusion algorithm, to validate the accuracy of cloud-covered LST reconstruction.

IV. Results and Analysis

A. Validation of Accuracy Improvement of Spatiotemporal Synergistic Temporal LST Interpolation Optimization

The fitting results indicate that the JNG06-STCI model achieves a marked improvement in fitting accuracy for cloud-covered pixels compared with the traditional JNG06 model.

From the temporal perspective, as shown in Fig. 6: Fig. 6(a) illustrates that for pixels nearly entirely covered by clouds throughout the day, the improved model enhances the computational capability for data-insufficient pixels by filling multiple data points during the sunrise and peak periods, thereby increasing the number of valid input data points beyond the theoretical minimum required for DTC model fitting (6 points); Fig. 6(b) demonstrates that under the condition of missing sunrise-period data, the improved model exhibits a marked reduction in fitting deviation, with the maximum deviation from the adjacent clear-sky DTC fitting results decreasing from 6.42 to 0.9 K; Fig. 6(c) shows that under the condition of missing data during the LST peak period, the model fitting deviation is also improved, with the maximum deviation reduced from 22.89 to 1.27 K.

From a spatial distribution perspective, the JNG06-STCI model exhibits optimization effects during the daytime, particularly during sunrise (Fig. 7) and the peak (Fig. 8) periods. Figs. 7(a)–(c) and 8(a)–(c) provide a side-by-side comparison:

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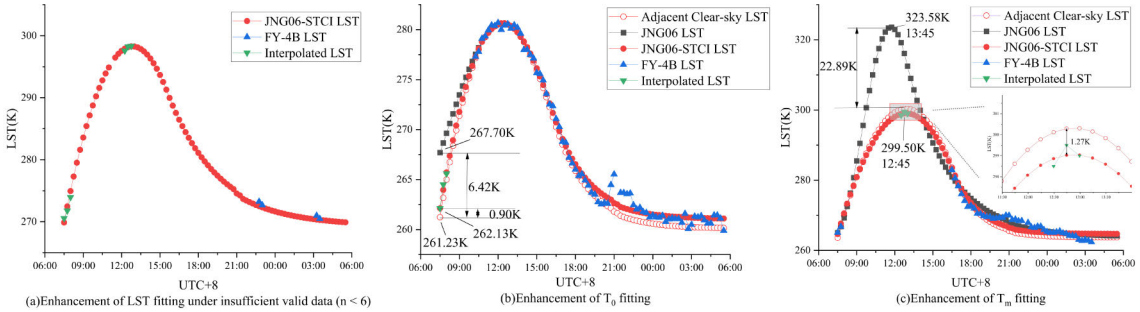


Fig. 6. JNG06-STCI (red) versus JNG06 (black) fitting results for three cloud-affected pixels. (a) Fewer than six valid observations. (b) Missing sunrise-period observations. (c) Missing peak-period observations. Blue: FY-4B clear-sky observations; green: interpolated T_0 and T_m; and hollow red: neighboring clear-sky pixel fits.

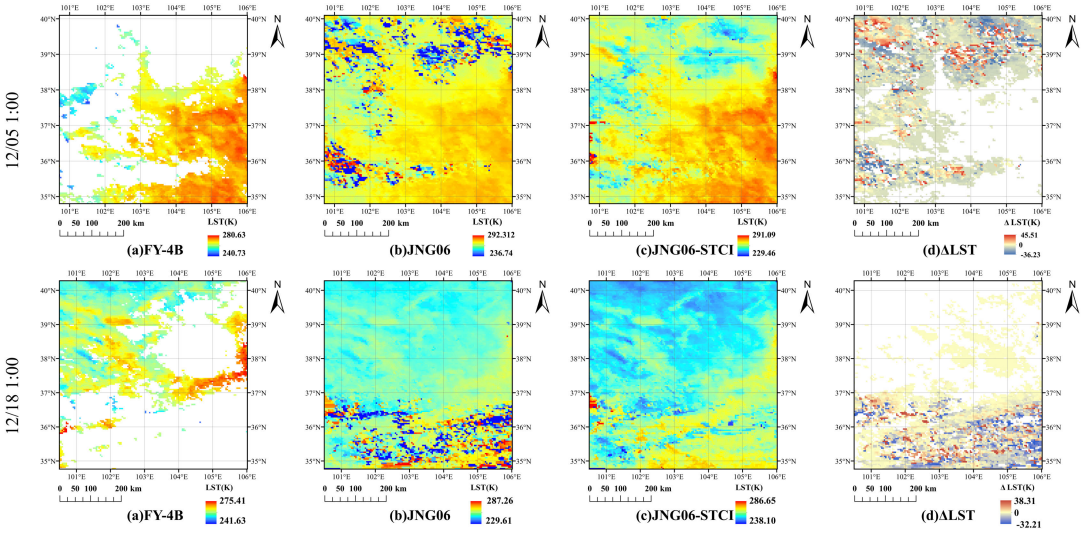


Fig. 7. Spatial LST comparison at sunrise on representative cloudy days. (a) FY-4B clear-sky retrieval (white: cloud-covered pixels). (b) JNG06 fit. (c) JNG06-STCI fit. (d) Difference between JNG06-fit and JNG06-STCI-fit (white: clear-sky pixels). Color bars: LST (K) in (a)–(c); difference (K) in (d).

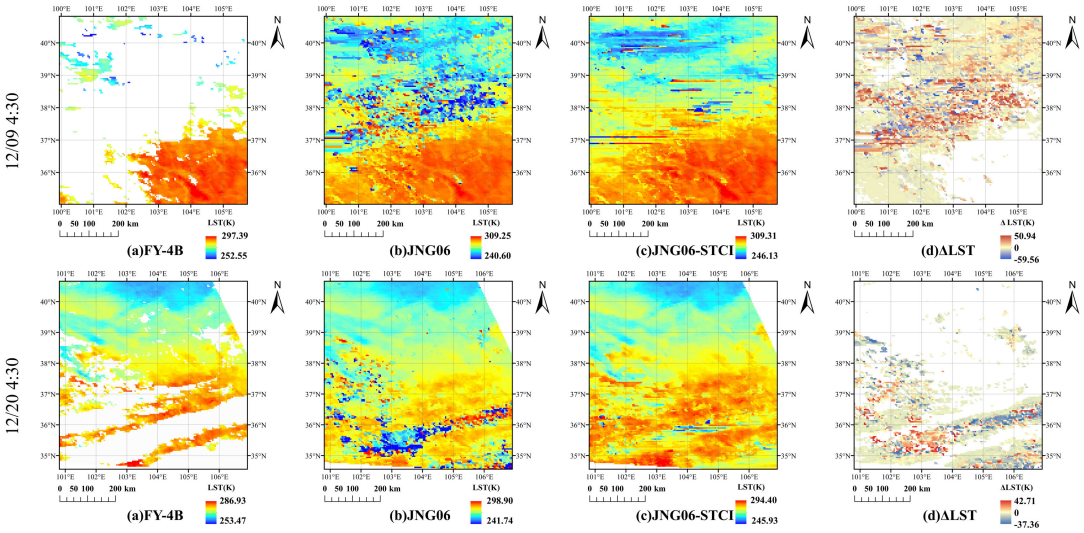


Fig. 8. Spatial LST comparison at local noon on representative cloudy days. (a) FY-4B clear-sky retrieval (white: cloud-covered pixels). (b) JNG06 fit. (c) JNG06-STCI fit. (d) Difference between JNG06-fit and JNG06-STCI-fit (white: clear-sky pixels). Color bars: LST (K) in (a)–(c); difference (K) in (d).

(a) the FY-4B clear-sky observation, (b) the JNG06 fit, and (c) the JNG06-STCI fit. Relative to JNG06, JNG06-STCI suppresses LST noise, enhances the spatial continuity of the LST field across cloud–clear boundaries, and constrains

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TABLE IV
SPATIAL SMOOTHNESS METRICS COMPARISON BETWEEN JNG06 ANDJNG06-STCI MODELS

Metric	JNG06	JNG06-STCI	Reduction(%)
Gradient magnitude Std (K/pixel)	4.05	2.40	40.8
Edge ΔT Mean (K)	3.42	2.30	32.8
Edge ΔT Std (K)	4.15	2.46	40.7
Semivariance at lag 1 (K^2)	6.73	2.68	60.1
Semivariance at lag 5 (K^2)	13.23	6.47	51.1

TABLE V
LST RMSE (K) BY LAND COVER TYPE AT DIFFERENT TIME PERIODS

Land Cover Type	Sample Size	All day		Sunrise to Noon		Noon to Sunset		Night	
		JNG06	JNG06-STCI	JNG06	JNG06-STCI	JNG06	JNG06-STCI	JNG06	JNG06-STCI
Evergreen Needleleaf Forests	12905	1.09	0.96	1.50	1.26	0.81	0.80	0.77	0.77
Open Shrublands	12772	0.61	0.61	0.35	0.36	0.60	0.59	0.72	0.72
Savannas	12660	0.89	0.90	0.63	0.65	0.88	0.84	1.01	1.03
Grasslands	12838	0.66	0.65	0.60	0.59	0.89	0.88	0.95	0.95
Permanent Wetlands	12932	0.87	0.85	0.72	0.80	0.72	0.61	0.81	0.83
Croplands	12716	0.61	0.62	0.45	0.46	0.73	0.73	0.66	0.66
Urban and Built-up Lands	12851	0.60	0.60	0.42	0.42	0.78	0.78	0.65	0.65
Permanent Snow and Ice	12639	0.11	0.12	0.06	0.05	0.14	0.15	0.09	0.09
Barren	12749	0.61	0.61	0.50	0.50	0.55	0.61	0.92	0.92

abnormally extreme values within the study area, so that the resulting LST range remains consistent with the physical characteristics of typical terrestrial surfaces. Figs. 7(d) and 8(d) make this improvement explicit by directly mapping the pixel-wise difference between the two fits. The discrepancy grows progressively from the cloud–clear boundary toward the interior of the cloudy region. This pattern mirrors the spatial distribution of cloud-cover duration: pixels deep inside the cloudy region are typically those obscured for the longest continuous period and thus retain the fewest valid observations for model fitting. In short, the magnitude of the JNG06-STCI correction is spatially consistent with the degree of JNG06 distortion: the more severely a pixel is cloud-covered and the less reliable the original fit, the larger the correction applied.

To quantitatively evaluate the improvement of spatial continuity introduced by the JNG06-STCI model, two complementary spatial smoothness analyses are carried out at different scales.

First, five scene-level metrics are computed over all valid pixels and averaged across all dates and time slots, characterizing spatial smoothness from three perspectives: local heterogeneity (standard deviation of the LST gradient magnitude); edge discontinuity (mean and standard deviation of the cross-boundary LST difference, edge ΔT Mean and Std); and short- to medium-range coherence (semivariance at lags of 1 and 5 pixels). For all five, smaller values indicate smoother fields. As summarized in Table IV, the gradient magnitude Std is reduced by 40.8%, the boundary discontinuity by 32.8% and 40.7%, and the lag-1 and lag-5 semivariances by 60.1% and 51.1%. This confirms that JNG06-STCI suppresses abrupt jumps at the cloud–clear boundary while also reducing local roughness and restoring spatial coherence across the field.

Second, to isolate reconstruction quality within the cloud-covered region, empirical semivariograms are computed over cloud-covered pixels only, for eight consecutive daytime hours on December 18, 2023. As shown in Fig. 9, at every hour and lag, the JNG06-STCI semivariance values are consistently lower than those of JNG06: the JNG06 curves rise steeply to sills of 35–65 K^2, whereas JNG06-STCI rises gradually and stabilizes at 10–20 K^2. The consistently lower and flatter semivariograms indicate that JNG06-STCI yields a hypothetical clear-sky LST with substantially improved spatial continuity within the cloud-covered region.

In summary, by interpolating the critical sunrise and peak observations and constraining the diurnal-spatial trend, JNG06-STCI corrects the fitting distortion that JNG06 suffers under data scarcity—reducing the maximum deviation from neighboring clear-sky fits to within ~1 K—and lowers all scene-level smoothness metrics by 32.8%–60.1%. The resulting hypothetical clear-sky baseline regains spatial continuity consistent with the surrounding clear-sky observations, providing a reliable input for the subsequent fusion.

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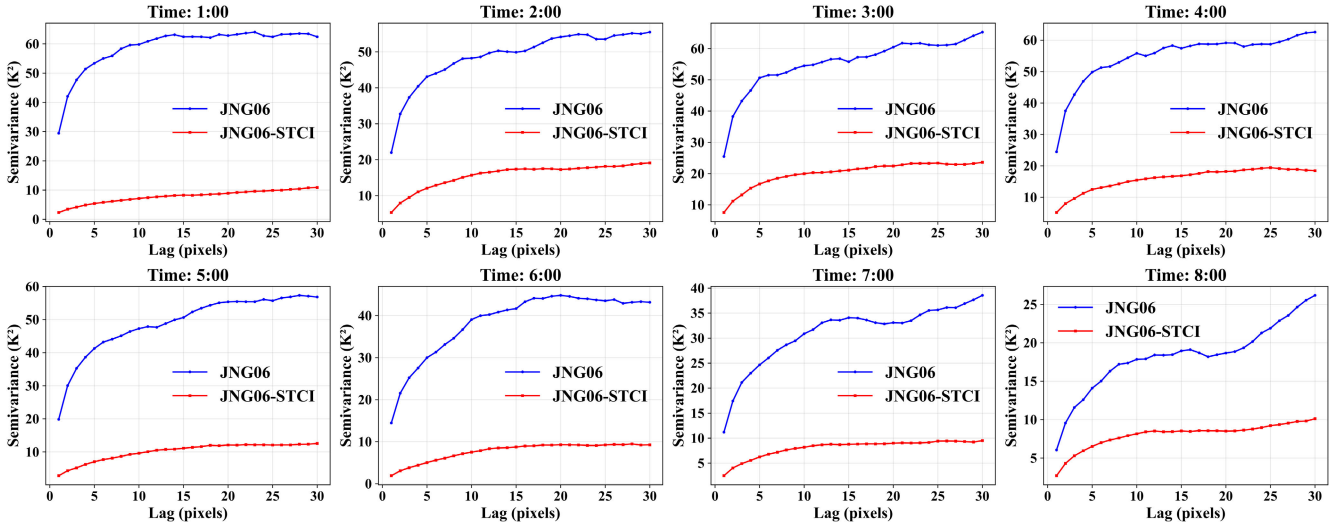


Fig. 9. Empirical semivariograms computed exclusively over cloud-covered pixels for the JNG06 (blue) and JNG06-STCI (red) models at eight consecutive UTC hours (01:00–08:00, corresponding to local daytime 09:00–16:00 BJT) on December 18, 2023, over the Heihe River Basin. The horizontal axis denotes the spatial lag distance (pixels; 1 pixel ≈ 4 km) and the vertical axis denotes the semivariance (K^2).

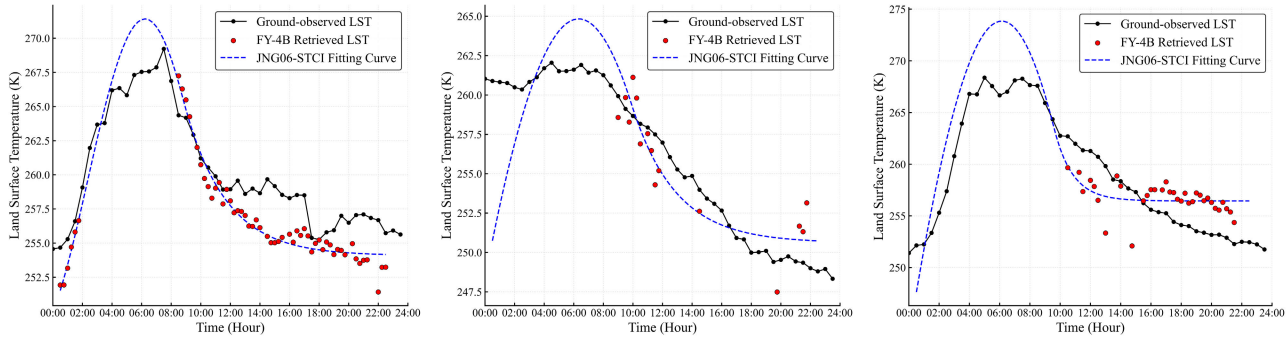


Fig. 10. Diurnal LST curves at three HiWATER stations on cloudy days. (a) Daman (cropland), December 17, 2023. (b) Desert (barren), December 14, 2023. (c) Heihe Remote Sensing (grassland), December 19, 2023.

B. Validation of Fitting Accuracy for the DTC Model

Several common land cover types in the study area were selected to verify the accuracy of the JNG06-STCI model under clear-sky conditions, with results presented in Table V. The overall fitting accuracy of the JNG06-STCI model under clear-sky conditions is comparable to that of the JNG06 model, with the all-day RMSE remaining below 1 K, demonstrating the model's robust baseline performance across different land cover types and providing a reliable clear-sky reference for subsequent cloud-covered LST reconstruction. In terms of diurnal phases, the highest accuracy is observed during the rapid warming period from sunrise to noon (UTC + 8 08:30–13:00), with a mean RMSE of 0.56 K; the secondary accuracy occurs during the sinusoidal cooling period from noon to sunset (UTC + 8 13:00–15:30), with a mean RMSE of 0.66 K; and the lowest accuracy is found during the nocturnal freedecay period (UTC + 8 15:30–until the next sunrise), with a mean RMSE of 0.73 K. Across land cover types, permanent snow/ice shows the highest fitting accuracy, with a daily mean RMSE of 0.12 K, whereas evergreen needleleaf forest exhibits the lowest accuracy, with a daily mean RMSE of 0.96 K; the accuracy for all other surface types remains between 0.61 and 0.90 K. These results demonstrate that the JNG06-STCI model can accurately capture the diurnal dynamics of LST under clear-sky conditions across different land cover types. Its stable and consistent fitting performance establishes a reliable initial data foundation for high-confidence reconstruction of LST in cloud-covered pixels.

C. Departure of the JNG06-STCI Baseline From Actual Cloud-Covered LST: Necessity of Reanalysis-Based Energy Balance Correction

After spatiotemporal interpolation, JNG06-STCI yields a spatially coherent diurnal curve and accurately captures clear-sky LST evolution. However, this fit only represents the hypothetical clear-sky LST and necessarily diverges from the observed cloud-covered LST whenever cloud radiative forcing perturbs the surface energy balance. This section illustrates that deviation using ground measurements and motivates the need for an additional energy-balance correction.

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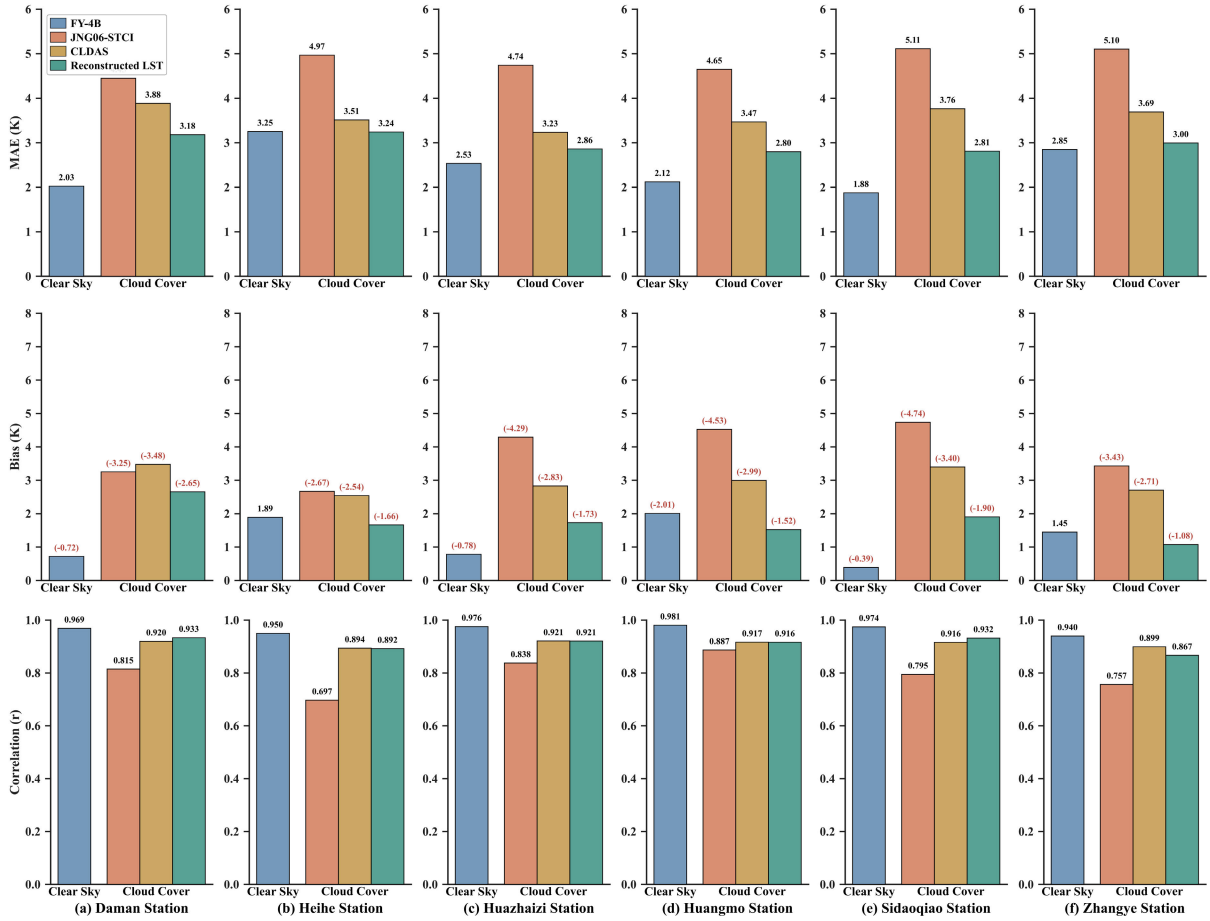


Fig. 11. Performance evaluation: FY-4B retrieval accuracy under clear-sky conditions compared to JNG06-STCI gap-filling, CLDAS retrieval, and the proposed algorithm's reconstruction accuracy under cloud-covered conditions. (a) Daman station. (b) Heihe station. (c) Huazhaizi station. (d) Huangmo station. (e) Sidaoqiao station. (f) Zhangye station.

TABLE VI
VALIDATION RESULTS (RMSE) USING GROUND-BASED STATION DATA

Station	Clear Sky	Cloud Cover			
	LST/K	JNG06-STCI/K	CLDAS/K	Proposed fusion/K	Sample Size
Desert Station	2.78	5.71	4.29	3.28	88
Sidaoqiao Station	2.25	7.70	4.74	3.24	87
Daman Station	2.45	5.78	4.92	3.73	187
Heihe Remote Sensing Station	3.95	6.93	4.29	3.79	115
Huazhaizi Desert Steppe Station	2.95	6.16	4.08	3.35	135
Zhangye Wetland Station	3.39	7.41	4.84	3.66	81

Fig. 10 compares the ground-measured LST, the FY-4B clear-sky retrievals, and the JNG06-STCI fitted diurnal curve over full 24-h cycles at three HiWATER stations representing distinct cloud regimes:

Fig. 10(a) shows Daman Station (cropland) on December 17, 2023. During the peak-temperature phase, the observed LST curve is notably flatter than the JNG06-STCI baseline. Rather than exhibiting the typical sinusoidal rise-and-fall, LST follows a subdued, plateau-like trajectory, indicating that sustained cloud radiative shielding prevents the surface from reaching the values predicted by the clear-sky DTC model.

Fig. 10(b) shows the Desert Station (barren) on December 14, 2023. The departure from the idealized DTC shape is more pronounced: because the site remains cloud-covered for most of the day, the observed LST changes little from sunrise to noon and decreases continuously from noon through the following sunrise, bearing no resemblance to the sinusoidal daytime warming pattern assumed by the DTC model.

Fig. 10(c) shows Heihe Remote Sensing Station (grassland) on December 19, 2023. The situation resembles that in Fig. 10(a): the observed LST tracks the baseline during the early morning but diverges progressively as cloud influence intensifies around midday.

Across all three cases, the in situ LST evolution deviates from the theoretical diurnal trajectory prescribed by the DTC

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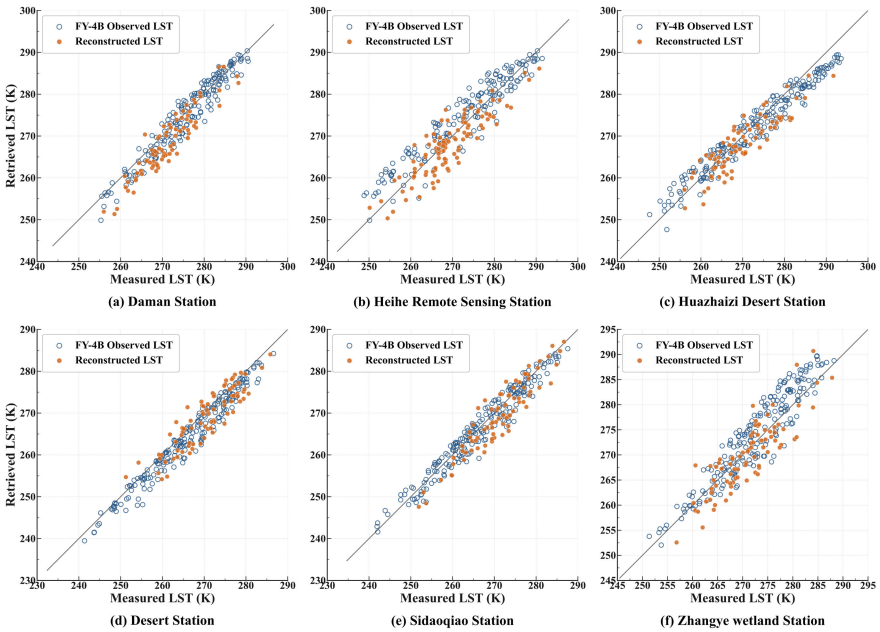


Fig. 12. Scatter plots of satellite-derived versus in situ LST at six HiWATER stations in December 2023. Hollow blue: clear-sky FY-4B retrievals; solid orange: cloud-covered reconstructed values. Solid line: 1:1 reference. (a) Daman station. (b) Heihe remote sensing station. (c) Huazhaizi Desert Station. (d) Desert station. (e) Sidaoqiao station. (f) Zhangye Wetland Station.

model, confirming that the JNG06-STCI baseline alone cannot represent the actual LST under cloud cover.

D. Validation of Cloud-Covered LST Reconstruction Accuracy

Table VI summarizes the station-level RMSE of four LST estimates against in situ measurements: 1) the FY-4B clear-sky retrieval; 2) the JNG06-STCI hypothetical clear-sky LST applied without reanalysis correction; 3) the raw CLDAS LST; and 4) the proposed fusion product.

Under cloud cover, the JNG06-STCI output employed in isolation yields a mean RMSE of 6.68 K (5.71–7.70 K). As a hypothetical clear-sky baseline, it does not account for cloud-induced alterations of the surface energy balance and therefore systematically overestimates the actual cloud-covered LST. The raw CLDAS LST is capable of characterizing surface energy balance processes and yields a mean RMSE of 4.53 K; however, its accuracy is still constrained by the systematic interdataset bias.

By integrating the spatially continuous JNG06-STCI baseline with the CLDAS-derived energy balance correction, the proposed framework reduces the mean RMSE to 3.52 K—improvements of 47.3% and 22.3% over the standalone JNG06-STCI and CLDAS results, respectively—and closely approaches the clear-sky retrieval accuracy of 2.99 K, with stable performance (3.24–3.79 K). The contrasts 2) versus 4) and 3) versus 4) confirm that the energy balance correction and the systematic bias correction are both indispensable to the framework.

Fig. 11 presents the station-level MAE, Bias, and correlation coefficient (r) for the four estimates. The proposed fusion product consistently surpasses both the standalone JNG06-STCI baseline and the raw CLDAS LST in MAE and Bias, and approaches the FY-4B clear-sky retrieval. The improvement is clearest in Bias: the JNG06-STCI baseline systematically overestimates the cloud-covered LST (up to about 4.5 K at the desert sites), whereas the fusion product reduces the bias to within roughly 1.5 K across all stations. For r, the fusion product (≈0.90) is comparable to CLDAS (0.91) and well above the JNG06-STCI baseline (0.75).

Fig. 12 presents the residual distributions between clear sky observed LST and reconstructed cloud-covered LST for each station. The similar distribution patterns and dispersion levels indicate that the data fusion reconstruction algorithm introduced in this study does not introduce additional errors or systematic biases when generating cloud-covered LST, and its error characteristics remain statistically consistent with those of clear-sky observations. This confirms that the reconstruction accuracy remains close to that of clear-sky observation.

Fig. 13 presents the all-weather cloud-covered LST reconstruction product proposed in this study, which integrates the LST results of the JNG06-STCI and CLDAS reanalysis LST data. This product achieves high-accuracy hourly LST reconstruction with a spatial resolution of 4 km. In the product, the LST data observed by the FY-4B satellite are adopted for clear-sky areas, while the cloud-covered areas are reconstructed based on CLDAS data corrected by the XGBoost algorithm. Compared with the hypothetical clear-sky LST results fitted by the JNG06-STCI model, the fused product after energy balance correction exhibits a more significant LST gradient difference between cloud-covered and clear-sky areas. The overall LST distribution in cloud-covered areas tends to be lower, which is consistent with the expected behavior of surface energy balance.

In summary, the proposed multisource data fusion approach effectively recovers cloud-covered LST and provides a robust technical pathway for spatiotemporally continuous regional LST estimation.

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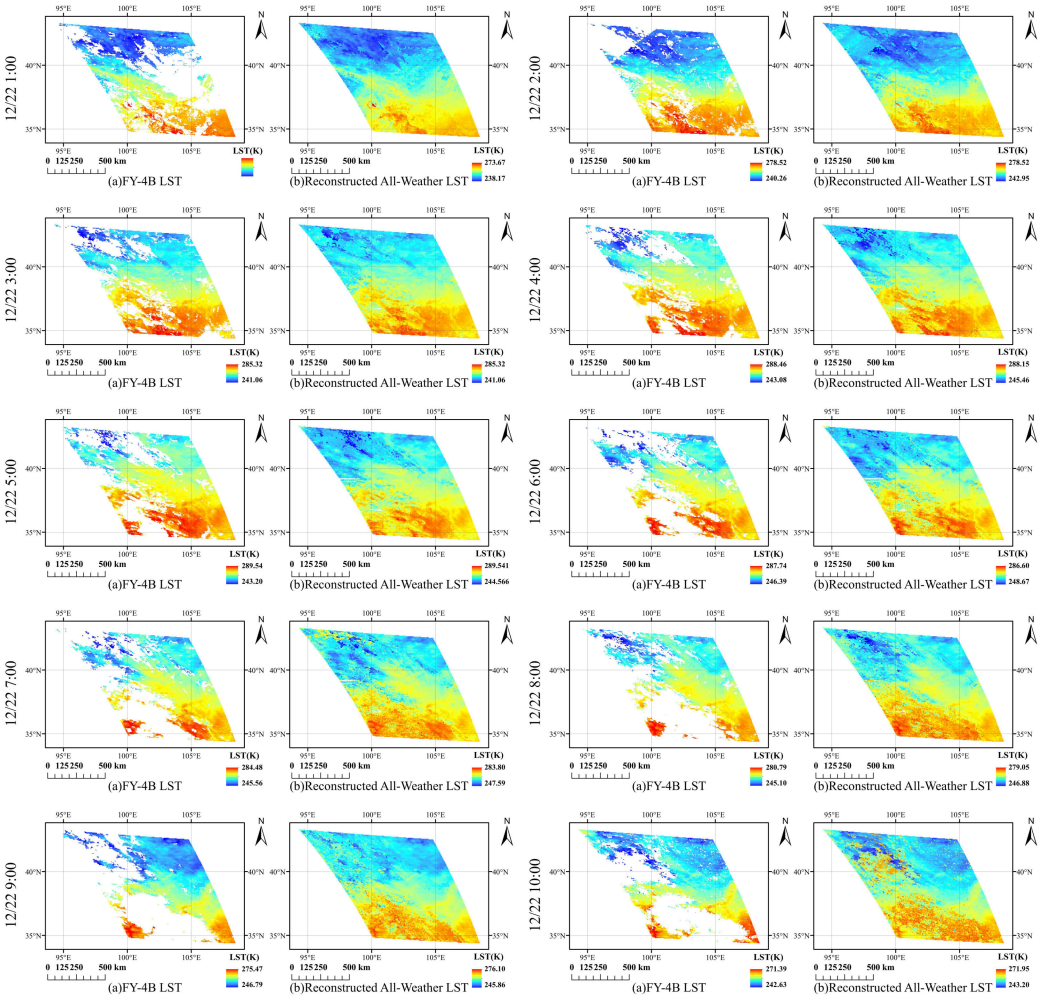


Fig. 13. Reconstructed all-weather hourly LST over the Heihe River Basin for a representative day in December 2023 (4 km, 1 h). Clear-sky pixels: FY-4B retrievals; cloud-covered pixels: fusion reconstruction. (a) FY-4B LST. (b) Reconstructed all-weather LST.

V. Discussion

A. Application Value of the All-Weather LST Product

The reconstructed FY-4B LST product provides hourly, 4-km LST under both clear-sky and cloud-covered pixels, and this spatial continuity is useful in several applications. In drought and agricultural water-stress monitoring, indices such as the temperature–vegetation dryness index (TVDI) and the temperature condition index (TCI) require gap-free LST; cloud-covered reconstruction extends these indices to overcast periods that clear-sky data alone cannot cover [58]. In surface urban heat island (SUHI) analysis, clear-sky-only sampling biases observations toward fair-weather conditions and overestimates SUHI intensity in humid cities [59]; incorporating cloud-covered LST yields a less biased estimate of the diurnal SUHI cycle [60]. In soil moisture downscaling, LST is a primary covariate for refining SMAP, SMOS, and ERA5-Land products from 9 to 25 km to 1 km, and a spatially continuous all-weather LST gives the downscaled fields near-daily coverage [61]. In cryospheric and permafrost monitoring over high-elevation and high-latitude regions, the product supplies more complete inputs to freeze–thaw indices and permafrost thermal models [62]. In each case, the reconstructed product complements existing clear-sky LST datasets by filling their cloud-induced gaps in space and time.

B. Limitations and Future Prospects

Despite the accuracy gains reported above, the proposed algorithm still has two limitations that warrant further work.

1) *Incorporating Additional Surface Energy Balance-Related Variables:* The fusion algorithm is based on the surface energy balance theory, but currently only incorporates surface radiation-related variables, namely surface NSR and upward/downward longwave radiation. At night, the absence of solar radiation drives NSR to zero, removing a key model input and degrading nighttime fitting accuracy. Introducing additional energy-balance variables—such as ground heat flux, soil moisture (available from CLDAS), and surface type—would enrich the inputs and improve reconstruction at night and under complex conditions.
2) *Integrating Passive Microwave Observations:* For long-term and large-scale continuous cloud cover exceeding three days, the DTC-based spatiotemporal interpolation

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loses the alternative LST observations it depends on, and fitting errors grow. Passive microwave radiation penetrates clouds and atmospheric water vapor, so it can retrieve LST under conditions where thermal infrared fails. Incorporating passive microwave data as an additional information source would extend the method to such persistent overcast and, given the higher revisit frequency of microwave sensors, supply observations over otherwise data-void regions [57], [63], [64].

VI. Conclusion

To address the issue of missing cloud-covered LST data in thermal infrared remote sensing, this study proposes a novel algorithm based on multisource data. The JNG06-STCI model is developed, with FY-4B temporal LST input data optimized through spatiotemporal interpolation strategies. In addition, a normalization model integrating FY-4B DTC-fitted data and CLDAS LST data is constructed based on the XGBoost algorithm to invert the actual cloud-covered LST data, thereby enhancing the accuracy and reliability of cloud-covered LST reconstruction. The main conclusions are as follows.

The JNG06-STCI model is developed based on the traditional JNG06 model by introducing a hierarchical spatiotemporal synergistic interpolation strategy. This approach enables the fitted hypothetical clear-sky LST dataset for cloud-covered pixels to exhibit spatial continuity consistent with clear-sky observations, effectively resolving the fitting distortions of the traditional DTC model in cloud-covered regions and filling the data gaps caused by cloud cover. The spatially seamless hypothetical clear-sky LST dataset generated by this model provides a spatially seamless input for subsequent data fusion algorithms.

A normalization model based on surface energy balance theory is constructed with the XGBoost algorithm, fusing FY-4B DTC-fitted LST with CLDAS reanalysis data. Rather than applying a single lumped correction, the satellite–reanalysis bias is explicitly decomposed into a clear-sky systematic discrepancy and a cloud-induced energy balance deviation, each modeled from independent training data and distinct physical drivers.

Validated against six HiWATER stations, the reconstructed cloud-covered LST achieves a mean RMSE of 3.52 K, improving on the standalone JNG06-STCI baseline (6.68 K) and raw CLDAS (4.53 K) by 47.3% and 22.3%, respectively, and closely approaching the clear-sky retrieval accuracy of 2.99 K. The framework delivers an hourly, 4-km all-weather LST product that fills the cloud-induced gaps that conventional thermal infrared products cannot resolve.

Future work will incorporate additional energy-balance variables (e.g., ground heat flux and soil moisture) to strengthen nighttime estimation, and integrate passive microwave observations to extend the method to persistent multiday cloud cover. By explicitly separating the systematic and cloud-induced bias components, the proposed framework reconstructs cloud-covered LST at near clear-sky accuracy and offers a practical route to spatiotemporally continuous, all-weather LST monitoring.

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